

# Radu Briciu

BSc, PGDip

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## EDUCATION

### Bayes Business School (formerly Cass)

City, University of London — Postgraduate Diploma

City of London, United Kingdom

Sept. 2023 – Jun. 2025

### Westminster Business School

University of Westminster — Bachelor of Science (with honours)

Westminster, London, United Kingdom

Aug. 2019 – Jun. 2023

## EXPERIENCE

### Finance & Operations Intern

Novo Nordisk UK Ltd

London, United Kingdom

Aug. 2021 – Aug. 2022

- Completed a 12-month placement split between the ‘Commercial Excellence’ and ‘Business Finance Analytics and Pricing’ teams
- Built ‘Sales vs Target’ and cost variance models and automated periodic reporting tasks
- Collected, synthesized, and presented analytics on market dynamics, brand performance, and customer identification
- Verified and consolidated financial accounts and supported analysis of financial performance
- Cleaned CRM data and collected competitor price data

### Intern

Bristol-Myers Squibb

Bucharest, Romania

Jun. – Jul. 2018

- Managed databases, documented archives, and performed basic data analysis
- Consolidated notes and planning documentation

## PROJECTS

### Machine Learning Option Pricing

[github.com/boomelage/machine-learning-option-pricing](https://github.com/boomelage/machine-learning-option-pricing)

Deep neural networks pricing barrier and Asian options on Heston-calibrated SPX surfaces, trained on large simulated grids and benchmarked against QuantLib; basis of the publication below.

### Heston Simulation

[github.com/boomelage/heston-sim](https://github.com/boomelage/heston-sim)

Semi-analytical Monte Carlo scheme for the Heston stochastic volatility model, using exact non-central chi-squared variance sampling and a trapezoidal integrated-variance update, with a vectorized NumPy implementation.

### Binomial Option Pricing

[github.com/boomelage/bopm](https://github.com/boomelage/bopm)

Derivation of a closed-form expression for the Cox–Ross–Rubinstein binomial model, collapsing the backward lattice recursion into a single discounted binomial sum over terminal payoffs.

### Realised Volatility

[github.com/boomelage/realised-volatility](https://github.com/boomelage/realised-volatility)

A statistical enquiry into the measurement of realised volatility, fitting ARMA–GARCH models to high-frequency returns and comparing forecast accuracy across specifications.

### Efficient Frontier Optimization

[github.com/boomelage/ef-optimization](https://github.com/boomelage/ef-optimization)

Markowitz mean–variance portfolio allocation in Python with constrained quadratic optimization and efficient-frontier visualization across the risk–return trade-off.

### prettymath

[github.com/boomelage/prettymath](https://github.com/boomelage/prettymath)

L<sup>A</sup>T<sub>E</sub>X package wrapping theorem-like statements in colour-coded, cross-referenced boxes, with configurable environments for definitions, theorems, and proofs in mathematical notes.

### Web Tetris

[free-tetris.online](https://free-tetris.online)

Browser Tetris with levels, line-clear scoring, piece hold, and ghost preview, written in dependency-free vanilla HTML/CSS/JavaScript.

## PUBLICATIONS

Briciu, Radu (Sept. 2024). “Neural Networks for Exotic Option Pricing”. DOI: [10.2139/ssrn.5104328](https://doi.org/10.2139/ssrn.5104328).

## COURSES

Advanced Corporate Finance – Applied Machine Learning – Asset Pricing – Blockchain, Cryptocurrencies and Decentralized Finance – Corporate Finance – Corporate Financial Accounting – Derivatives – Development Economics – Econometrics of Financial Markets – Financial Modelling and Statistics – Fixed Income – Foundations of Econometrics – Global Business Environment – Introduction to Finance – Introduction to Financial Markets – Investments and Portfolio Management – Law and the Business – Mathematics for Finance – Money Banking and Financial Markets – Numerical Methods: Applications – Polylang German 1 – Professional and Research Skills for Finance – Programming for Quants – Risk Analysis – Risk Management – Securities Analysis – Stochastic Modelling Methods in Finance – Strategic Perspectives for Finance and Accounting

## SKILLS

Machine Learning –  Python (QuantLib, NumPy, Pandas, sklearn, keras) – MATLAB – L<sup>A</sup>T<sub>E</sub>X – CLI (PowerShell, Bash, WSL, Git, Github) – Editors (Sublime Text, Vim, VSCode, T<sub>E</sub>Xstudio, JupyterLab) – CLI Agents (Claude Code, Codex, Gemini) – Ollama – BIOS/OS Configuration – HTML/CSS/JavaScript – MS Office & Power Query